Chapter 7

Methods of Exploring Protein–Ligand Interactions to Guide Medicinal Chemistry Efforts

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Abstract

We present a number of techniques to analyze protein–ligand interactions in the context of medicinal chemistry: crystal Contract Preferences, Electrostatic Maps and pharmacophore screening using Hückel Theory. Contact Preferences is a statistical technique to predict hydrophobic and hydrophilic geometry in receptor active sites. Electrostatic Maps use the Poisson-Boltzmann Equation to model solvation effects and are particularly useful for predicting hydrophobic regions. Pharmacophore annotation with Hückel Theory provides finer detail of hydrogen bonding interactions, including CH..O interactions. Applications to AblK:Gleevec and CDK2 virtual screening are presented.

Key words Molecular interactions, Contact statistics, Electrostatic maps, Pharmacophore annotation

1 Introduction

The combination of the widespread availability of X-ray crystal structure coordinates of macromolecular structures (e.g., as found in the Protein Data Bank [1] and affordable 3D computer graphics hardware) has made it possible for scientists to routinely visualize such structures in their efforts to design better inhibitors and drugs.

While visually inspecting atomic coordinates and bonding patterns can be revealing, it is often not easy to perceive structures such as the shape of a receptor's active site. A molecular surface is drawn about the receptor (e.g., a Connolly surface [2] can provide such information in a readily discernable manner). Similarly, a molecular surface constructed from an iso-contour of the van der Waals energy between a receptor and a hypothetical "probe" atom can highlight sterically accessible regions in an active site. Such surfaces are often useful when contemplating modifications to a ligand in order to improve activity or selectivity. Receptor properties such as atom type, electrostatic energy, and hydrogen bond information can be used to color code the molecular surfaces in an effort to

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further assist with the interpretation of the macromolecular structure information.

In medicinal chemistry applications, crystal structures of protein–ligand complexes are used to understand important interactions, such as hydrogen bonds, halogen bonds, ionic and hydrophobic interactions, in an effort to guide ligand modification for improved potency, selectivity, or physical properties. Hydrogen bonding interactions tend to be directional, localized and can be modulated by ligand substituents, especially the weaker CH...O interactions. Hydrophobic and ionic interactions tend to be more diffuse and can be affected by solvation/cavitation effects, which present challenges when attempting to specify a pharmacophore for a particular active site.

In this chapter, we will examine a number of techniques of determining and analyzing non-bonded interactions between proteins and ligands with the intended application of guiding ligand design. This guidance takes the form of understanding existing interactions, predicting potential new or modulated interactions and the specification of pharmacophores for virtual screening.

2 Contact Preferences

The x-ray crystal structures in the Protein Databank and the Cambridge Structural Database [3] are a good source of non-bonded contact information from which statistics can be derived. Methodologies such as X-Site [4] and SuperStar [5] are knowledge-based techniques that use statistical distributions derived from crystallographic data to describe or predict non-bonded contacts between ligands and proteins. The main idea is to describe (statistically) the geometry of non-bonded interactions and use the statistical descriptions to predict the likelihood of interactions in a specific protein active site. Knowledge-based methods are appealing because they rely on experimental data and not molecular mechanics, precise hydrogen coordinates, or other approximations; however, they do rely on sufficient experimental data and appropriate a priori classification of interacting atom types (e.g., sp^2 donor vs. sp^3 acceptor).

We will describe one example of a statistical method, *Contact Preferences* [6], for determining geometric preferences for polar and hydrophobic atoms given the 3D coordinates of a protein receptor. Fundamentally, Contact Preference maps are contours of probability densities indicating a percentage likelihood of a non-bonded contact between a protein receptor and a particular ligand atom type; in other words, the likelihood that a given contact geometry would be observed in the crystallographic databases. Interactions in high-probability regions (e.g., donor and acceptor pair at ideal geometry) are considered good and interactions in low probability

regions (e.g., hydrophobic atoms covering a hydrogen bond position) are considered non-optimal.

We are interested in the probability density

$$\Pr(l, x|s)$$

the probability of observing a ligand atom of type l at position x in contact with a structure s consisting of a collection of atoms with types $\{t_i\}$. We approximate.

$$\Pr(l, x \mid s) = \Pr(x \mid l, s)\Pr(l \mid s)$$

$$\approx c \max_{i} \Pr(x \mid l, t_i)\Pr(l \mid t_{i.})$$

where c is a normalization constant and have assumed that we can compose the distribution over the entire structure from the individual receptor atoms.

The $\Pr(l \mid t_i)$ term (the probability that a ligand type l is in contact with a receptor type, t_i) is estimated straightforwardly using relative frequencies of atomic contacts in a training set; for example, the relative frequency that the (l, t) pair is observed within 4.5 Å over sidechain-sidechain contacts in the PDB.

Two vectors **u** and **v** (*see* Fig. 1) are used to define a coordinate system about each receptor atom located at Cartesian coordinates, y, depending on hybridization and heavy atom coordination number. In some cases **v** is taken to be 0 (e.g., 3-coordinated sp^2 atoms). For an interacting atom located at x, we define the spherical-type coordinates

$$r = |x - y| \quad a = \frac{180}{\pi} cos^{-1}\left[\boldsymbol{u}\cdot\frac{x-y}{r}\right] \quad p = \frac{180}{\pi} sin^{-1}\left[\boldsymbol{v}\cdot\frac{x-y}{r}\right]$$

where r is the distance between l and t_i, a is the angle relative to **u** in the plane normal to $\boldsymbol{v}$, and p is angle relative to **u** in the **uv** plane. Assuming independence of the individual coordinates we approximate

$$\Pr(x \mid l, t_i) \approx \Pr(r \mid l, t_i)\ \Pr(a \mid l, t_i)\ \Pr(p \mid l, t_i)$$

and estimate the individual one-dimensional probability densities from a collection of sidechain-sidechain (l, t) contacts found in the PDB.

For each ligand receptor atom type pair, histograms are collected over the PDB and these histograms are fitted with analytical distributions. For example, Fig. 2 shows the histograms and fitted distributions for the receptor atom type "oQ1," an sp^2 oxygen with one heavy atom neighbor. One can clearly see the directional preference for the in-plane hydrogen bonding interactions at lone pair directions ($a = 60°$) and the out-of-plane hydrophobic interactions ($p > 45°$).

Figure 3 depicts the 95% probability contour levels for a collection of receptor atom types created from the analytical probability

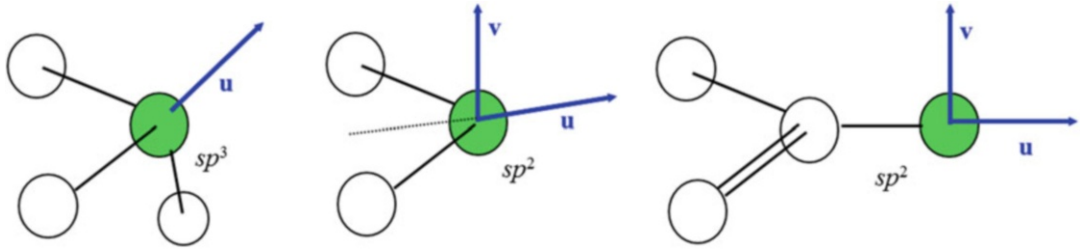


Fig. 1 Two vectors **u** and **v** centered on a receptor atom (*green*) define a local coordinate system; the directions of these vectors depend on the receptor atom's hybridization and heavy coordination number

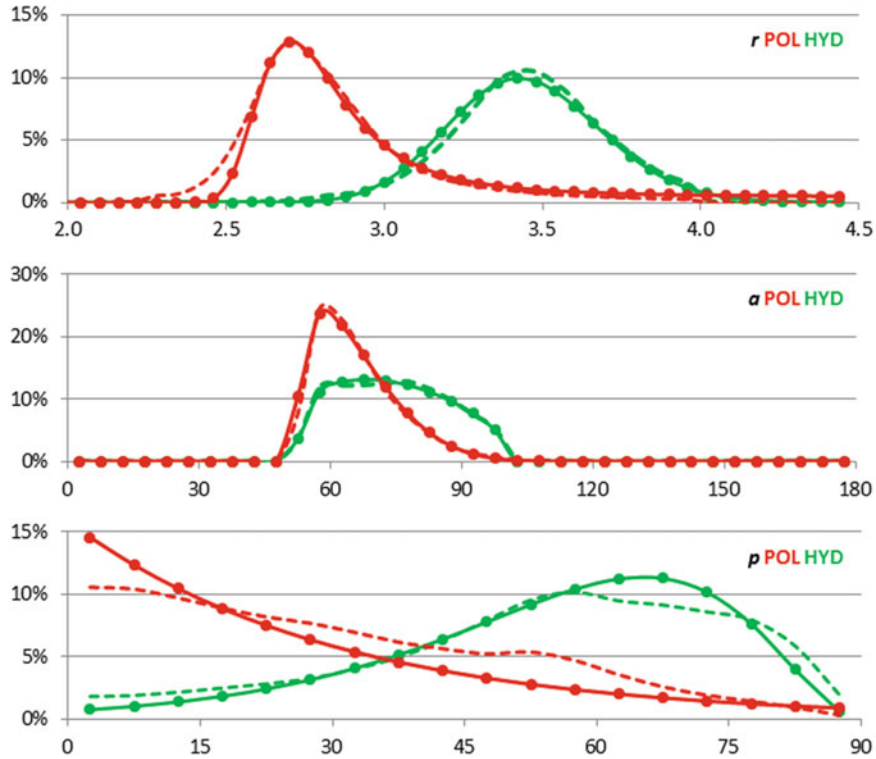


Fig. 2 Distributions of non-bonded contact coordinates between ligand atom types polar (POL in red) and hydrophobic (HYD in *green*) and a receptor atom of type "oQ1", sp^2 oxygen with one heavy neighbor. The *dashed lines* are the histograms collection from sidechain-sidechain contacts in the PDB and the *solid lines* are analytical fitted distributions. The top distribution is for the inter-atom distance, the middle the in-plane angle, and the bottom is the out-of-plane angle (*see* the text)

distributions fitted to PDB statistics. Receptor atom types are denoted by their element (lower case for sp^2) and heavy atom neighbor count. Ligand atom types are either polar (POL) or hydrophobe (HYD). Hydrogens are displayed for reference; however, these were not used in the statistical fit, which is why, for example, the primary amine type "NQ1" does not show directionality with respect to the neighboring hydrogens.

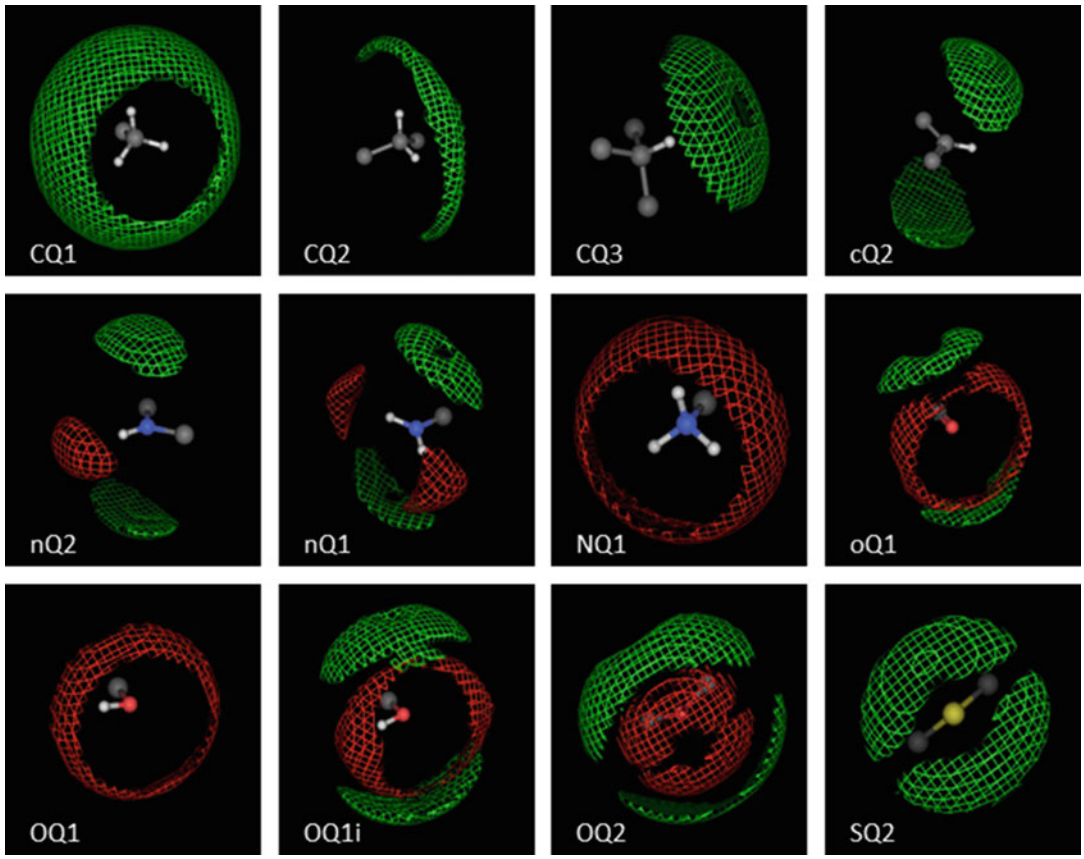


Fig. 3 95% probability contour levels for interactions with a receptor atom type and ligand atom types POL (*red*) and HYD (*green*)

There are a number of uses for such probability contours. Composite contour level plots in receptor active sites can indicate regions of high probability of polar or hydrophobic contacts which can be useful in structure-based design. The directional preferences are also useful for annotating ligand atoms' hypothetical contact partners in pharmacophore 3D search techniques. Also, the statistical distributions can be used to assess the geometry quality of hydrogen bonds in graphical interaction displays.

However, there are notable drawbacks to statistical techniques. First, the method is sensitive to the a priori atom typing which causes a loss of detail of chemical context; for example, phenol oxygens (type "OQ1i") are generally good hydrogen bond acceptors except for weakening due to substituent effects in ligands. The statistics are gathered for *all* phenol oxygens which conflate the chemical contexts leading to potentially misleading contact probabilities. Second, because hydrogen atoms are generally not visible in the PDB there is loss of distinction between donor and acceptor partners, for example, with hydroxyl oxygens. Third, each receptor

atom is treated independently; consequently, compositing the maps in a receptor active site neglects correlation and longer range electrostatic effects. Fourth, the probability maps reflect, in some sense, only the frequency of occurrence of contacts in crystal structures and not necessarily a strength or energy value.

3 Electrostatic Maps

Molecular mechanics potentials are one method of addressing some of the deficiencies of statistical approaches. Forcefield terms such as Coulomb or Reaction Field electrostatics and Lennard-Jones van der Waals energies are physically motivated and rely on only a few parameters per atom. In principle, distinct chemical contexts can be better described and interaction correlation effects can be better treated (in contrast to the independent-atom assumption of statistical approaches).

An early example of the use of physical potentials to identify favorable binding sites in a receptor is Goodford's GRID [7, 8] in which iso-energetic surfaces of the van der Waals, electrostatic and hydrogen bond energy between a small probe atom (or fragment) and a macromolecular structure are used to identify favorable binding sites. The surfaces are displayed on a graphics terminal and the binding regions in space are identified by suitably adjusting the energy values used to produce the contour surfaces.

Originally, a Coulomb potential was used; however, this gas-phase model overemphasized the electrostatics interactions. Moreover, hydrophobic interactions were difficult to quantify due to the lack of an implicit solvent model. Subsequently, an implicit solvent model was incorporated based upon spatial differences in the dielectric of solvent versus solute. Notwithstanding the fact that the implicit solvent model took only the macromolecular structure and a particular location of the probe into account, the GRID methodology has found utility in efforts to understand ligand binding [9–11]

A more sophisticated treatment of macromolecular electrostatics uses the Poisson–Boltzmann Equation (PBE) which is a partial differential equation used in the study of electrolytes. Methods to solve the PBE have been the subject of much research [12–14] because of their ability to model implicit solvent and buffer effects. Typically, a macromolecular structure is used to define a region of low dielectric (e.g., 1) in a certain volume and the remaining points of space are assigned a high dielectric (e.g., 80). The solution to the PBE has largely been used to predict pK_a values for ionizable groups or to color code molecular surfaces. Such color coding can be revealing since it presents a self-consistent view of the electrostatic field near a solute with solvent screening taken into account.

It is tempting to use the PBE electrostatic field in GRID-type methodology; however, the application of a single solution is flawed because the electrostatic field depends on the low dielectric cavity which depends on all particles, including the probe. Either a prohibitively long calculation would be required (one PBE solution for every lattice value calculated) or the cavitation effects of the probe would have to be neglected.

An alternative, called Electrostatic Maps [15], uses solutions of the PBE to augment the GRID-type methodology. The starting point is the Poisson equation which is a fundamental equation of physics that relates the equilibrium electrostatic potential φ at a point in space and a charge density ρ at that point:

$$\nabla \cdot \nabla \varphi + \rho = 0$$

where $\nabla \cdot \nabla = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$ is the Laplacian operator. Given a charge density, ρ, one can calculate the electrostatic field φ by solving the partial differential equation. In the special case of a collection of point charges, φ is the familiar Coulomb potential.

Suppose that a charge density (from a protein or small molecule) ρ is interacting with a number of mobile ions (e.g., from salts) each carrying a formal charge, q_I. At thermodynamic equilibrium these mobile ions will be distributed about ρ according to a charge density which we will denote by ρ_I. The electrostatic potential φ from the combined charge densities must satisfy the Poisson equation

$$\nabla \cdot \nabla \varphi + \rho_I + \rho = 0$$

For multiple types of mobile ions, the foregoing equation is easily extended by replacing the lone ρ_I with a sum of similar densities (one for each type). For the purposes of exposition, only one mobile ion type will be considered.

If the mobile ions are spatially restricted (e.g., the ions cannot be inside a protein and must remain in solvent), then such a restriction can be specified with a spatial potential u_I that has high energy in the forbidden regions of space and low energy in the allowed regions of space. Taken with the electrostatic potential φ the potential energy of a mobile ion at a particular point in space will be, at equilibrium, $q_I\varphi + u_I\, q_I$. We now assume that the ion location density, ρ_I, follows a Boltzmann distribution and so the ion charge density is its formal charge multiplied by the probability it will be located at a particular point in space

$$\rho_I(x) = q_I c\, e^{-[q_I\varphi(x)+u_I(x)]/kT}$$

where c is normalization constant, k is Boltzmann's constant, and T is the temperature of the system. If φ and u_I are zero sufficiently far from ρ then at such distances the exponential term tends to one which means that c may be taken as the bulk concentration, C_I, of the mobile ions (e.g., in mol/L). The combination of the previous

two equations results in a Poisson-Boltzmann type equation, namely that

$$\nabla\cdot\nabla\varphi + q_I C_I\, e^{-[q_I\varphi(x)+u_I(x)]/kT} + \rho = 0$$

which specifies not only the resulting electrostatic potential, but also the ionic charge densities both of which result from an original charge density ρ.

The Electrostatic Maps methodology consists of the display of the equilibrium potential energy iso-surfaces of mobile "oxygen" and "hydrogen" particles carrying partial charges q_O and q_H, respectively, and subject to spatial Lennard-Jones van der Waals potentials u_O and u_H, respectively. Upon (numerical) solution of the above PBE, we have that

$$\nabla\cdot\nabla\varphi + Cq_O e^{-\left[q_O\varphi+u_O\right]/kT} + 2Cq_H e^{-[q_H\varphi+u_H]/kT} + \rho = 0$$

where C is the bulk concentration of water and ρ is the charge distribution of a macromolecule under consideration. The potential function $q_O\varphi + u_O$ is the potential energy landscape of the implicit "oxygen" particle and $q_H\varphi + u_H$ is the potential energy landscape of the implicit "hydrogen" particle. These two potentials represent the free energy minimizing energy landscapes of the neutralizing, or screening, solvent particles. In some sense these represent the energies required to displace solvent at particular points in space under the assumption that the remaining points in space are ideally (in a mean field sense) solvated or neutralized. An additional "hydrophobic" potential, $-(q_O + q_H)\varphi + u_C$, is defined (where u_C is a Lennard-Jones potential for carbon) which will have negative values in neutral regions (absence of "O" or "H" particles).

Fundamentally, Electrostatic Maps are plots of van der Waals and electrostatic energies of implicit particles and, as such, they are similar in spirit to the GRID methodology with the mobile particles playing the role of the "probes." The difference, however, is that a single, nonlinear, self-consistent calculation is performed in which each implicit mobile particle feels the effects of the other implicit mobile particles in order to better model screening effects. This nonlinearity and self-consistency ultimately manifests itself in a more localized distribution of implicit particle densities. Approximate, or even exact $\varepsilon = 80$ solvation models can fail to adequately screen chemical groups leading to electrostatic domination of a region because there is a nonlinear screening effect in the buffered solution that cannot be modeled by linear ε dielectric models.

A simple example will serve to illustrate this point. Figure 4 shows the contour plots of the $q_O\varphi + u_O$ and $q_H\varphi + u_H$ functions for two different solvent models. On the left, a linear screening dielectric of $\varepsilon = 80$ outside of a small molecule was used in the PBE without mobile particles. One can see the large extent of the blue (preference for positive) contour created by the carboxylate group.

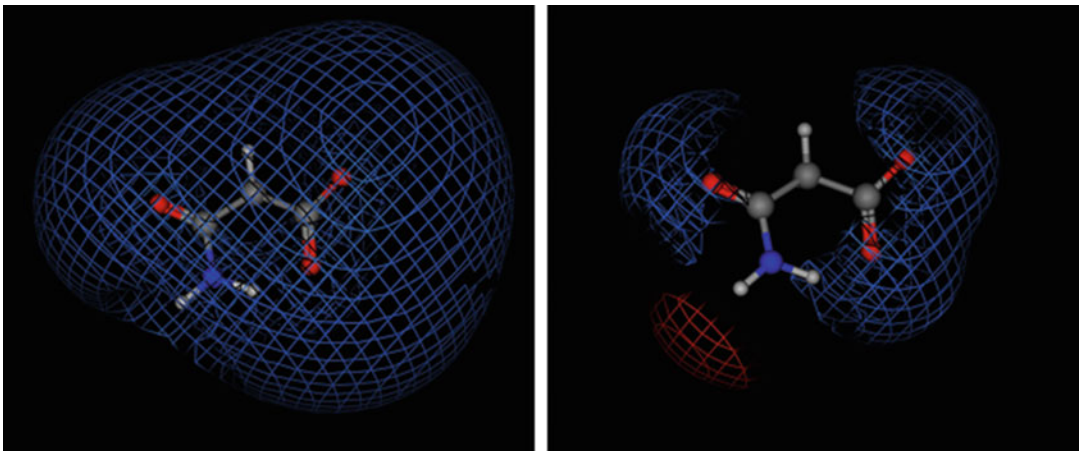


Fig. 4 (*Left*) Electrostatic and van der Waals energy contour plots of an anionic small molecule calculated by solving the Poisson equation in a polar medium ($\varepsilon = 80$). Positive preference is indicated in *blue* and negative preference is indicated in *red*. Notice that the carboxylate dominates the entire region and no red contour is visible. (*Right*) Electrostatic with van der Waals contour plots of the same small molecule calculated by solving the nonlinear PBE with "O" and "H" mobile screening particles (*see* the text). Notice that the screening effects are much stronger and that more localized positive and negative preferences are visible

On the right are the contours of the proposed Electrostatic Map calculation in which the contours are much more localized; so much so, that a preference for negative (red) appears next to the hydrogen in the amide group. Charged groups in aqueous environments are neutralized by the "oxygen" mobile counter-ions (modeled in Electrostatic Maps with the "oxygen" and "hydrogen" particles) that provide nonlinear screening effects and must be modeled using nonlinear Boltzmann term of the PBE. By using the mobile self-consistent Boltzmann particles of the PBE, the carboxylate group will be sufficiently screened so that groups beyond its first solvation shell will not feel an unrealistically strong electrostatic field.

We shall now use the AblK:Gleevec complex to show how Electrostatic Maps can be used to rationalize the activity trends in a ligand series. The Bcr-Abl chimeric protein has been implicated in the development of chronic myeloid leukemia (CML) and acute lymphoblastic leukemia [16, 17] Gleevec (Imatinib) is a small molecule inhibitor of the dephosphorylated (inactive) form of Bcr-Abl kinase currently being used clinically as a treatment for CML. Patients can develop resistance to Gleevec if Bcr-Abl expression is enhanced or if point mutations occur within the Abl kinase domain [17] and consequently there is interest in further developing next-generation Bcr-Abl kinase inhibitors.

It has been shown [18] that the Gleevec piperazine ring fits in the solvent exposed opening of the pocket and forms hydrogen bonds to Thr315, Ile360, Asp381, and Met318 (*see* Fig. 5). One

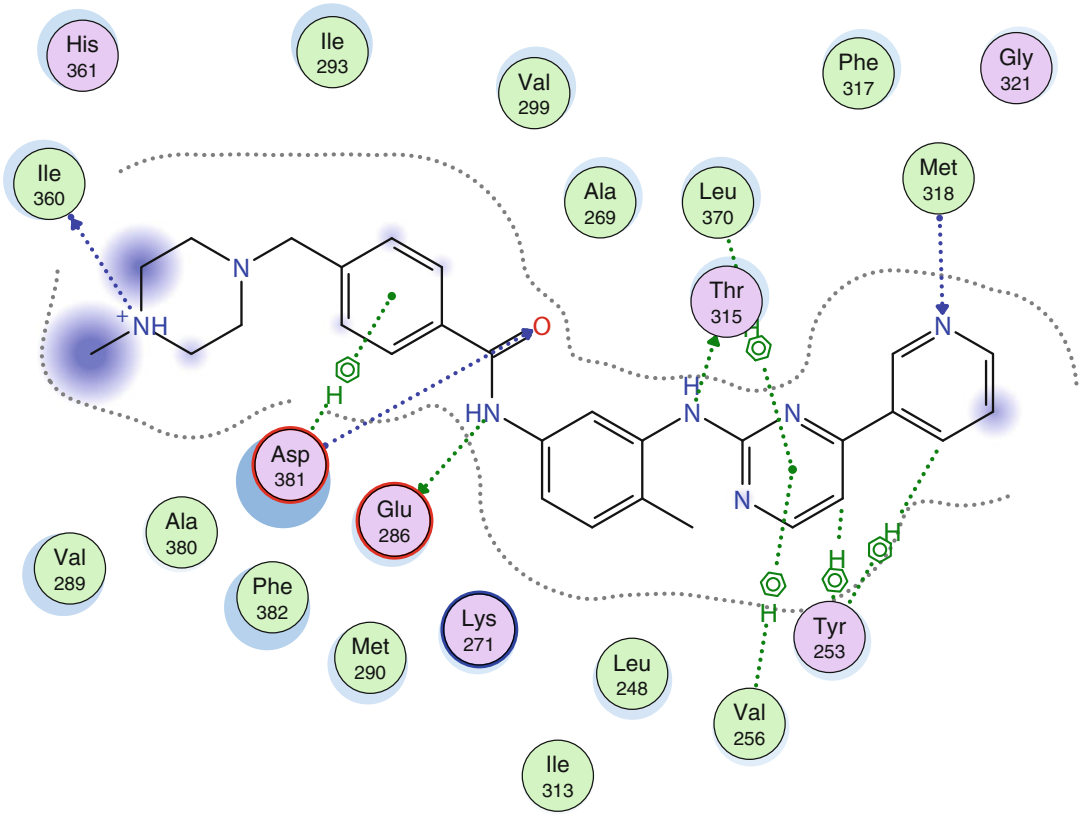


Fig. 5 A diagram of the interactions of Gleevec with Abl kinase in PDB:1IEP. Residues are represented by *discs* with polar residues in *pink* (acidic residues with a red contour and basic residues with a *blue contour*) and hydrophobic residues in *green*. *Dotted arrows* indicate hydrogen bonding to sidechain (*green*) and backbone (*blue*) atoms respectively. Blue "clouds" on ligand atoms indicate the solvent-exposed surface area of ligand atoms (larger means more exposure). Light-blue "halos" around residues indicate the degree of surface area contact with ligand atoms (larger means more contact). The dotted contour reflects steric room for methyl substitution.

would expect that an Electrostatic Map calculated from AblK alone to reproduce these interactions and possibly suggest regions of possible optimization.

The AblK protein was prepared for electrostatic analysis as follows. His361 was neutralized according to its immediate environment so that it could form hydrogen bonds to the backbone nitrogen of Asp363 and the backbone oxygen of Ala380. The remaining residues were assigned standard ionization states and protons were added in extended conformation. Partial charges were then calculated using the MMFF94 forcefield [19] and the electrostatic map was calculated from the receptor structure alone near the active site. Figure 6 shows the results of the calculation with the positive (blue) density plotted at −2.5 kcal/mol, the negative (red) density at −2.0 kcal/mol, and the hydrophobic (green) density at −3.0 kcal/mol. The calculated Electrostatic

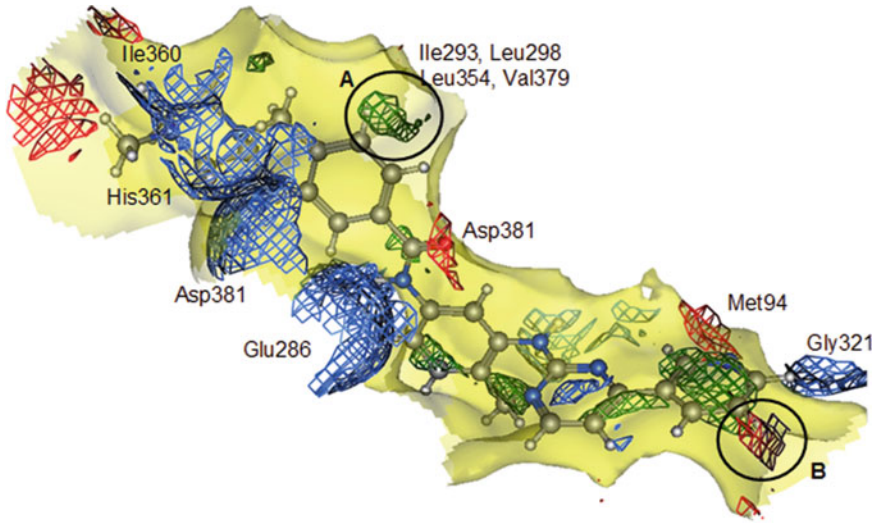
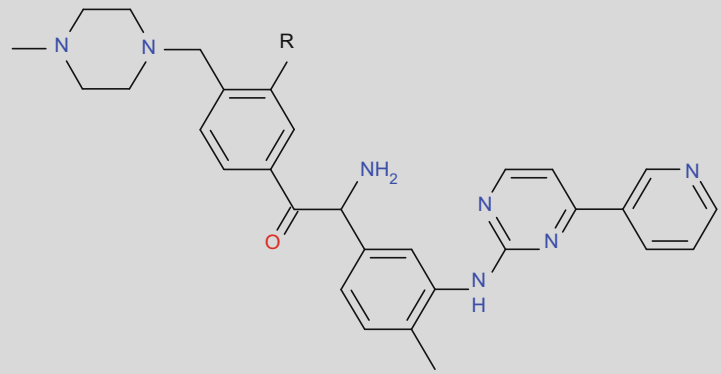


Fig. 6 The Electrostatic Map for the active site of AblK:Gleevec (1IEP) calculated from receptor atoms only; positive preference is indicated in *blue*, negative in *red*, and neutral in *green*. The *yellow* interaction surface shows the steric boundary of the pocket. *A* and *B* denote map densities not filled by Gleevec (*see* the text)

Map (*see* Fig. 6) agrees well with clear positive and negative charge densities corresponding to all of the important hydrogen bonding interactions of Gleevec. The predicted hydrophobic regions overlap well with both the methyl group attached to the phenyl linker as well as the pyridine ring fragment indicating strong favorable hydrophobic interactions.

There are two significant regions in the AblK Electrostatic Map that are not filled by Gleevec and therefore are potential regions of ligand optimization. The unfilled hydrophobic density (Fig. 6 region A) suggests that a small hydrophobic group such as a halogen, CH_3, or CF_3 would have favorable interactions with AblK and could be a location for optimization. This is in fact the case; Asaki et al. [18] reported biological activity data for a series of 3-substituted benzamide derivatives as Bcr-AblK inhibitors reproduced in Table 1; these compounds differ from Gleevec only by the phenyl ring substituent in the hydrophobic subpocket (created by Ile293, Leu298, Leu354 and Val379). The IC_{50} data of Table 1 shows that small hydrophobic substituents improve activity, in particular, the 3-trifluromethylated benzamide. The same 3-trifluoromethyl moiety is also present in NS-187 [20], a recently proposed Gleevec analog that not only binds more strongly to AblK than Gleevec, but maintains potency in the presence of many known point mutations. It is interesting that the gain in potency of approximately -2.1 kcal/mol $= -kT \ln (IC_{50}(CF_3)/IC_{50}(\text{Gleevec})$ of the CF_3 substituent is in rough agreement with the -3 kcal/mol hydrophobic contour value of the Electrostatic Map. This is most likely due to the fact that the Electrostatic Map (free) energies estimate the solvation effects that largely determine differences in affinity in this highly homologous series.

Table 1
Activity of Gleevec analogs against K562 cells



Compound	R	IC_{50} (nM)
Gleevec	H	183
1	F	63
2	Cl	10
3	Br	7
4	I	10
5	CF_3	5

The second unfilled region of the Electrostatic Map for AblK (Fig. 6 region B) suggests a preference for negative charge. This negative density (created by the hydroxyl of Tyr253) lies about 1.5 Å from the *meta* position of the pyridine ring at about 45° out-of-plane. Consequently, it is a difficult feature to reach. Asaki et al. [18] synthesized pyrimidine derivatives instead of the pyridine ring to compensate for the increased hydrophobicity caused by trifluoromethyl group addition to Gleevec and to avoid interfering with the Tyr253 (which stabilizes the inactive form of AblK) with bulky substitutions. Their results show no significant improvement in activity against Bcr-Abl K562 cell lines (estimated $\Delta\Delta G$ of 0.13 kcal/mol). This experimental conclusion could possibly have been elucidated by the analysis of the Electrostatic Map around the pyridine moiety because the negative (red) density is relatively far from the contemplated pyrimidine nitrogen. A contour level of −1 kcal/mol begins to show overlap with the contemplated nitrogen; however, since the hydroxyl and the nitrogen would be far from optimal hydrogen bonding geometry it is reasonable to conclude that an interaction would be weak, even if mildly favorable.

The Electrostatic Maps methodology has a number of advantages. The method is physically motivated and relies on molecular

mechanics parameters (partial charges and van der Waals parameters). It is fast to calculate, requiring a few seconds on common computer hardware. Correlation and solvation effects are taken into account in a self-consistent way (in contrast to statistical composites). The method is particularly useful for identifying important hydrophobic (electrostatically neutral) regions in an active site (e.g., for pharmacophore specification); this is in contrast to purely geometric approaches such as Contact Preferences which neglect longer range interactions. More sophisticated approaches such as 3D–RISM [21] have been the subject of much research and can deal with higher order correlations and cavitation effects. These techniques are much slower to calculate but can identify "non-standard" hydrophobic regions; for example, hydrophobic regions in the immediate vicinity of polar atoms created by the shape of an active site.

4 Pharmacophore Screening

The virtual screening of 3D conformation databases using a pharmacophore as a query is a widely used technique in computer-aided drug discovery, with hundreds of published accounts of its application since its advent in the late 1970s [22, 23] and the first published account of a successful virtual screen in 1992 [24]. Typically, a pharmacophore query (possibly derived from a receptor structure) specifies a collection of molecular features (hydrogen bond donor/acceptor, cation, anion, hydrophobe, etc.) intended to capture the essentials of ligand binding interactions to a receptor, along with inter-feature distance constraints. The query is applied to a database of small-molecule conformations to search for arrangements of features in 3D that match the pharmacophore query with respect to both feature type and distance constraints.

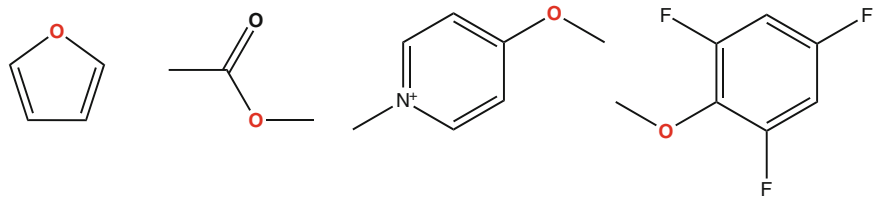
Commonly, features of ligands are assigned by substructure searching methods from a fixed template or rules collection. For example, a typical hydrogen bond donor heavy atom can be encoded with the SMARTS pattern "[#7,#8;!H0]" meaning "nitrogen or oxygen with at least one hydrogen." Assigning hydrogen bond acceptor features is not as straightforward—electron withdrawal and hyper-conjugation must be taken into account since, among others, conjugated ether oxygen and nitro oxygen are not good hydrogen bond acceptors [25]. The problem becomes particularly acute when one attempts to accurately encode CH hydrogen bond donors (e.g., in kinase inhibitors [26]). Electron withdrawal and substituent effects are very difficult to consistently encode in a (small) collection of substructure patterns.

An alternative to substructure patterns for assigning hydrogen bond donor and acceptor features is to use Hückel Theory quantum calculation to provide a consistent treatment of electronic effects. A small organic molecule is specified as a collection of

atoms, each with an atomic number, Z, a formal charge, Q, and a collection of bonds $\{(i,j)\}$ between the atoms. A self-consistent modified Hückel Theory calculation is performed on the chemical graph (without coordinates) using a well-known method [27]; the calculation produces a partial charge (or electron occupancy) per atom and a fractional π bond order per bond.

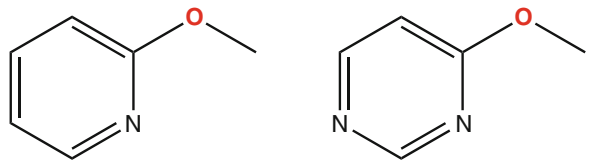
The partial charges and fractional bond orders are used to determine a hydrogen bond "donor strength," ε_{don}, and "acceptor strength," ε_{acc}, per atom using the methods described by Gerber [28] and are the key component of the non-bonded terms of the MAB forcefield [27]. In this forcefield, the energy of a hydrogen bond at ideal geometry is the product of the donor and acceptor strengths, $E_{\text{HB}} = \varepsilon_{\text{don}} \times \varepsilon_{\text{acc}}$; for example, an ideal water-water hydrogen bond has an energy of 1.56 kcal/mol. In the development of the strength coefficients, Gerber demonstrated their clear relationship with free energies of hydration.

It seems natural to use strength coefficient thresholds to define donors and acceptors in small molecules; in other words, a large enough ε_{acc} value would indicate an acceptor and a large enough ε_{don} value would indicate a donor. Using $\varepsilon_{\text{acc}} > 0.83$ (a hydrogen bond energy with water greater than 1 kcal/mol) we find that indicated oxygen atoms (in red) in the following structures

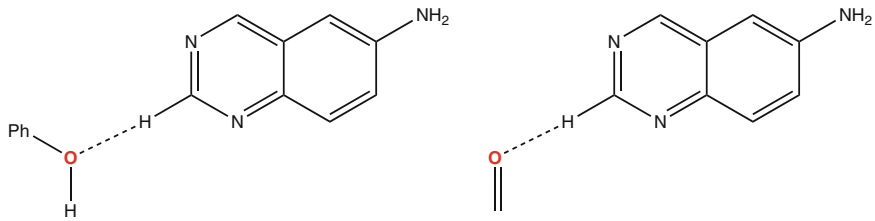


are not deemed acceptors, along with other similar cases that normally would have to be treated as special case exceptions in rule-based systems, while the carbon atoms in isonitrile and carbon monoxide were deemed acceptors (e.g., in metal ligation). Conventional alcohols, phenols, alphatic amines, etc. are properly annotated. Generally, there is very good agreement with hand curated acceptor rules. Using the same principle for the ε_{don} cutoff, we find very good agreement with hand curated rules. Alcohols, phenols, conjugated amines, conjugated thiols were deemed donors, while neutral aliphatic amines and aliphatic thiols were not deemed donors unless sufficient electron withdrawing groups were present (e.g., FCH_2SH)

Unfortunately, the strength threshold scheme does not produce satisfactory separation for CH donors: there does not appear to be any single a priori cutoff value that satisfactorily separates CH donors from non-CH donors, especially in substituted heterocycles. This is due to the fact that CH hydrogen bonds tend to be weak and have a rather continuous strength distribution. In fact a similar issue exists for borderline acceptors; for example, the following structures



straddle the $\varepsilon_{acc} > 0.83$ cutoff line, which in any case is somewhat arbitrary. These borderline examples highlight that what is required is to model the *interaction* between the small molecule and a hypothetical receptor in the pharmacophore query itself rather than an isolated small-molecule feature without context. In other words, the pharmacophore query should contain (hypothesized) information about the receptor's hydrogen bonding partner atom. For example, a strong acceptor in the receptor could match strong or weak donors in the ligand while a weak acceptor in the receptor should only match strong donors in the ligand.



A CH donor interacting with a Tyr OH (above) is an example of a weak acceptor—weak donor interaction with $\varepsilon_{don} = 0.57$ $\varepsilon_{acc} = 0.87$ and $E_{HB} = 0.50$ kcal/mol, whereas a backbone carbonyl oxygen produces a strong acceptor—weak donor interaction with $\varepsilon_{don} = 0.57$ $\varepsilon_{acc} = 2.00$, and $E_{HB} = 1.14$ kcal/mol. Clearly, the threshold comparison should take place on the hydrogen bond energy of the interaction and not the individual ligand strengths. In other words, a CH is a donor if it interacts with a suitably strong acceptor, otherwise it is not.

To make use of this concept, a pharmacophore query is augmented with a strength parameter on each hydrogen bond feature. In this way, the query becomes a kind of receptor model that contains inter-feature distance constraints, feature types, and a strength coefficient intended to model the nature of the receptor's corresponding hydrogen bond partner.

Figure 7 shows an augmented pharmacophore query; the augmented query (magenta sphere) contains information, either known or hypothesized, about the strength of the receptor's acceptor: $\varepsilon_{acc} = 2.1$ (a typical carbonyl oxygen value). The donor strength of the ligand is calculated as the search proceeds. If the geometric and feature type constraints are met, then the strength product is formed $0.8 \times 2.1 = 1.7$ kcal/mol. If this interaction energy is greater than a user-supplied threshold, say 1 kcal/mol, the structure is emitted as a hit, otherwise it is rejected.

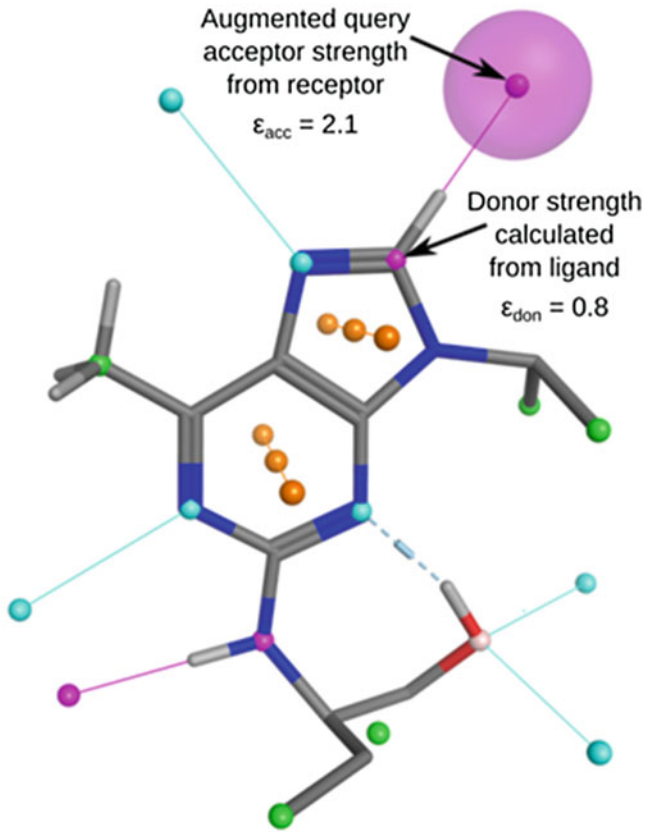


Fig. 7 The augmented pharmacophore query for CDK2 hinge binders with CH donors; the magenta sphere is a query feature with an encoded hydrogen bond acceptor strength of 2.1 of a hypothesized receptor atom. The small spheres represent potential pharmacophore features calculated from the ligand

CH donor interactions are often seen in ligands binding to the GluPheLeu hinge in kinases. Hinge binders form tridentate bond interactions with the backbone of the hinge binding in a similar fashion to ATP. We obtained the crystal coordinates of CDK2 in complex with pyrazolo[4,3-d]pyrimidine (PDB:3PJ8). The crystal structure was prepared with Protonate3D [29] and the coordinates refined. A pharmacophore query was constructed in an attempt to replace the pyrazolopyrimidine scaffold, while maintaining interactions with the hinge so that the substituent presentation vectors would be preserved. The query contained four features: two aromatic centroids, a donor, and an acceptor feature. The strength coefficient of the donor feature was set to 1.91 to model the acceptor strength of the carbonyl oxygen of the backbone of Glu81 in CDK2. Figure 8 (left) depicts the cocrystallized staring ligand in CDK2 along with the pharmacophore query.

We searched a 3D fragment database using the modified pharmacophore search (including the Hückel strengths) and the result was 66 hits that satisfied the pharmacophore query and the

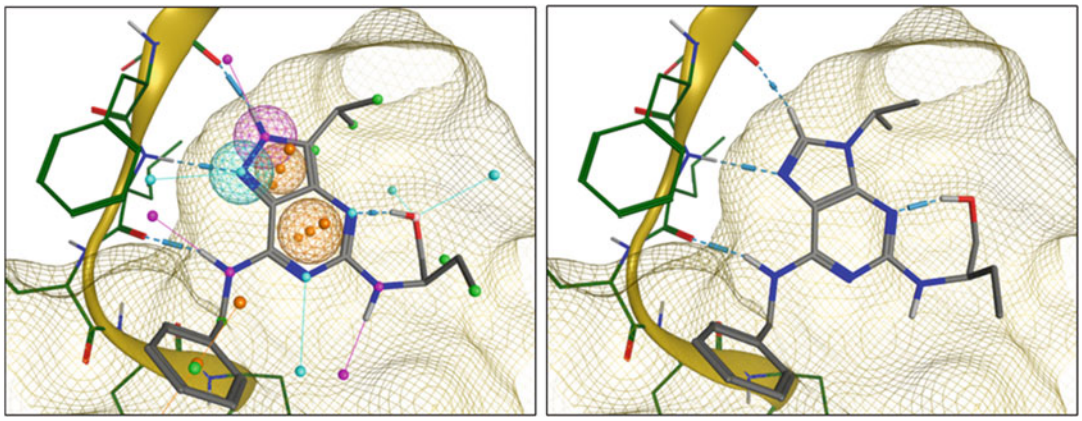
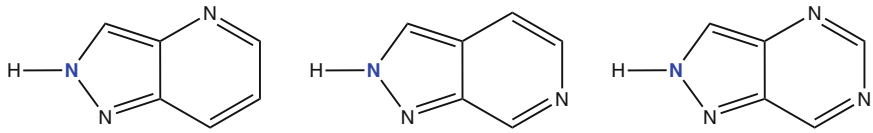


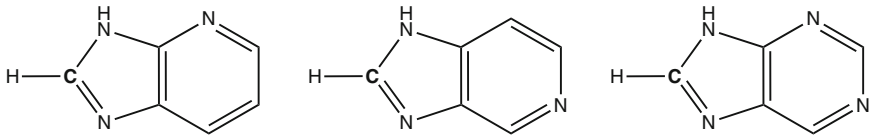
Fig. 8 *Left*: pyrazolo[4,3-d]pyrimidine cocrystallized with CDK2 (PDB:3PJ8) along with the four feature pharmacophore queries used in a scaffold replacement experiment. *Right*: top ranking CH donor imidazopyrimidine scaffold with attached substituents in a calculated binding mode with CDK2

substituent presentation vectors. Among the hits were conventional hydrogen bonding scaffolds (the parenthesized quantities are the Hückel hydrogen bond energies to the hinge, in kcal/mol):



pyrazolopyridine (6,67) pyrazolopyridine (6.69) pyrazolopyrimidine (6.88)

Note that the search recovered the original pyrazolo-pyrimidine scaffold. The search also retrieved scaffolds with CH donors that interacted with the hinge.



imidazopyridine (1.25) imidazopyridin e (1.21) imidazopyrimidine (1.34)

Note that the imidazopyrimidine framework, the top ranking CH scaffold, results in the compound seliciclib (purvalanol class of molecules); *see* Fig. 8 (right). A similar search using conventional pharmacophore typing and the identical pharmacophore query resulted in only 35 hits, none of which had the CH donor motif.

5 Conclusion

We have described a number of techniques to analyze protein–ligand interactions in the context of medicinal chemistry: crystal Contract Preferences, Electrostatic Maps, and pharmacophore screening using Hückel Theory.

Atom contact statistics extracted from crystal structures the PDB or the CSD can be used to describe and/or predict the directional preferences of hydrophobic or hydrophilic ligand atoms in receptor active sites. Methods such as Contract Preferences are appealing since they are largely based on an experimental observation. They are good at determining hydrogen bonding geometric preferences but poorer at identifying important hydrophobic regions in an active site. This failing is largely due to the neglect of long-range forces and correlation effects.

Methods based upon the Poisson-Boltzmann Equation such as Electrostatic Maps can capture long-range forces and solvation effects. These methods require assignment of hydrogen coordinates to crystal structures and atomic partial charges which capture local chemical context in ligands. The details of polar interactions are generally in broad agreement with statistical methods; however, important hydrophobic regions in an active site are more readily identified. An application to the AblK:Gleevec complex demonstrated the utility of the Electrostatic Maps method for structure-based design.

Fine detail of hydrogen bonding interactions, including the weaker CH donor interactions, requires a more sophisticated treatment. We presented a method based upon Hückel Theory to assign hydrogen bond strengths to protein and ligand atoms in the context of 3D pharmacophore search. An application to CDK2 was presented showing a replacement scaffold including a CH..O interaction could be determined with pharmacophore search.

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